Industrial Robot Configurations

Different types optimized for specific applications

Articulated (6−axis)

Most versatile
Human−like motion
6 degrees of freedom

Welding, painting assembly, handling

SCARA

Fast horizontal motion
Rigid vertical axis
4 axes typical

Assembly, pick−place packaging

Delta (Parallel)

Very high speed
Limited payload
Parallel linkage

High−speed picking packaging, assembly

Cartesian (Gantry)

Linear motion
Large work envelope
3−4 axes

CNC loading, palletizing large part handling

Cylindrical

Rotary base
Linear reach
4 axes typical

Machine tending material handling

Collaborative (Cobot)

Force limited
Safe around humans
Easy programming

Human collaboration assembly assist